



VMware Cloud
Foundation 9

Defending Against AI-Fueled Threats

with VMware Cloud Foundation 9

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Disclaimer

This document is intended to provide general guidance for organizations that are considering Broadcom solutions. The information contained in this document is for educational and informational purposes only.

This document is not intended to provide advice and is provided "AS IS."

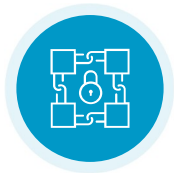
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Organizations should engage appropriate legal, business, technical, and audit expertise within their specific organization for review of requirements and effectiveness of implementations.

Why Folks Are Worried About AI



Finds & Exploits
Vulnerabilities



Chains
Vulnerabilities

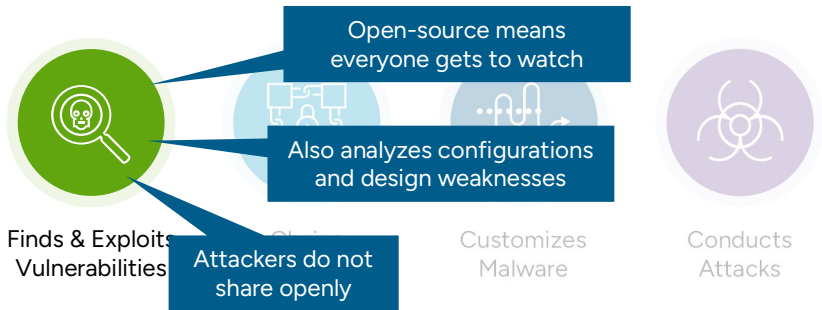


Customizes
Malware



Conducts
Attacks

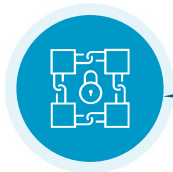
Why Folks Are Worried About AI



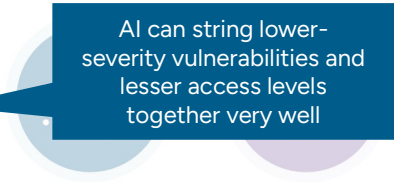
Why Folks Are Worried About AI



Finds & Exploits
Vulnerabilities



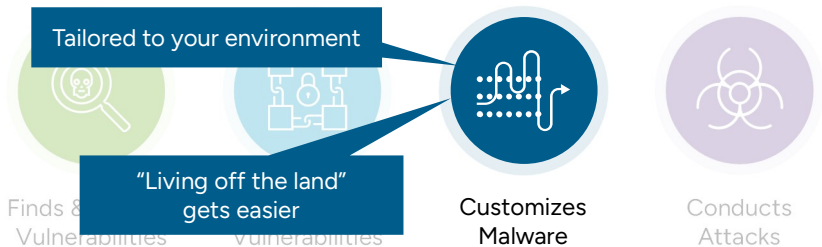
Chains
Vulnerabilities



Customizes
Malware

Conducts
Attacks

Why Folks Are Worried About AI



Why Folks Are Worried About AI



Organizations Need To Think Differently



Security Always

The perils AI brings to us turn many things into actual operational risk. Actual architectural zero-trust security practices need to be prioritized.



Patching Is One Piece

Patching is essential, but it is one piece among many. Identity, segmentation, logging, and recovery create resilience and defense-in-depth.



Survivorship Bias

A clean record is easily mistaken for a strong defense. Perhaps the test has not come, or an attacker is already inside, unnoticed.

Not Just Us Saying These Things

Different Audiences, Same Ideas

Cloud Security Alliance

Segmentation between systems

Identity isolation and phishing-resistant MFA

Deception and tripwires inside the network

Pre-authorized containment that runs at machine speed

FS-ISAC

Treat vulnerability backlogs as operational risk, not compliance debt

Shift from vulnerability management to exploit prevention

Move beyond CVSS-only prioritization

Replace end-of-life software, stay within two major versions

Anthropic

Patch internet-facing systems within 24 hours

Design for breach: hard barriers, not friction

Manual approval cycles are now a security risk

Identity as primary control, segmentation as backstop

What can be done?



Defend Holistically

Most intrusions exploit trust relationships between systems. Actual zero trust, plus strong identity and access controls help close gaps.



Limit Lateral Movement

Perimeter and endpoint defenses will fail eventually. Segmentation and least privilege turn a compromise into a contained problem.



De-Risk Patching

Snapshots, backups, canaries, and staged rollouts improve overall resilience (not just patching!) All features that are in or available for VCF.

SSO & Federation!
RBAC! Avi!
Log Management!



Defend Holistically

Most intrusions exploit trust relationships between systems. Actual zero trust, plus strong identity and access controls help close gaps.

W
be
vDefend!
VLANs! VPCs!
Distributed Switch!



Limit Lateral Movement

Perimeter and endpoint defenses will fail eventually. Segmentation and least privilege turn a compromise into a contained problem.

Protection & Recovery!
Replication! Avi!
Snapshots! Clones!



De-Risk Patching

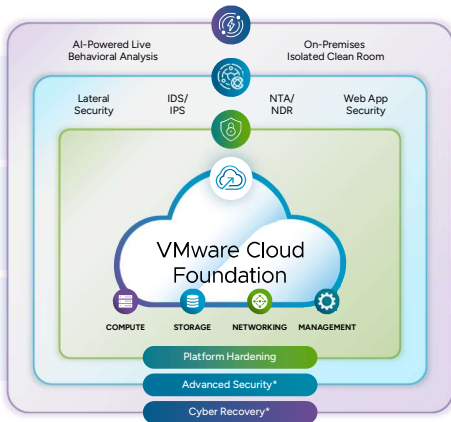
Snapshots, backups, canaries, and staged rollouts improve overall resilience (not just patching!) All features that are in or available for VCF.

Defense in Depth with VMware Cloud Foundation

Cyber-Resilient
Recovery

Strong Lateral
Security

Hardened
Infrastructure



- Continuous replication with rollback points
- Automated testing in isolated networks
- Choice of detection vendor
- **Faster recovery, verified clean**

- Visibility into every workload and flow
- Signature- and behavior-based detection
- Automated response and policy tuning
- **Reduce blast radius and dwell time**

- Compliance enforcement and drift detection
- Patching while workloads stay running
- High performance, secure storage
- **Resilient from hardware to workload**

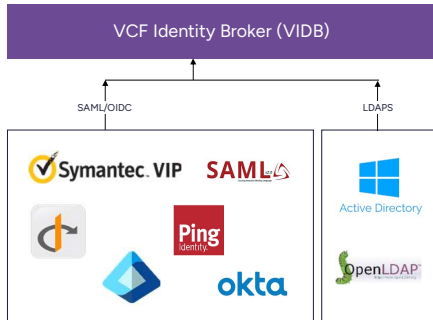
Make Identity Hard To Compromise

Authentication & Authorization



Unified Identity: One Broker for All VCF Components

VMware Cloud Foundation SSO



Embedded (in vCenter),
standalone, and clustered
appliance deployment models

Automatic configuration for
vCenter, NSX, Operations,
and Automation

One IdP configuration for VCF
management components, but
Automation has multitenancy

Supports any SAML, OIDC,
and AD/LDAPS-based IdP

Isolate Identity From Compromise

Authentication & Authorization with VMware Cloud Foundation 9

1. **Use VCF SSO to federate VCF management with an isolated, infrastructure-only IdP.**
2. **Enable phishing-resistant MFA, number matching, and conditional access.**
3. **Do not connect HR, user provisioning, password reset, or configuration management systems to the infrastructure IdP. Do not connect centralized monitoring systems to infrastructure in a way that lets them execute commands or push configurations.**
4. **Help Desk must not be able to reset administrator authenticators, including for remote staff. Reset in person for all privileged accounts. This is a common threat actor tactic now.**
5. **Lower authentication token lifetimes.**
6. **Reduce permissions for service accounts to the minimum needed.**
7. **Audit infrastructure IdPs against published best practices (disable SSPR, Seamless SSO).**

**"We cannot become what we want
by remaining what we are."**

- Max De Pree

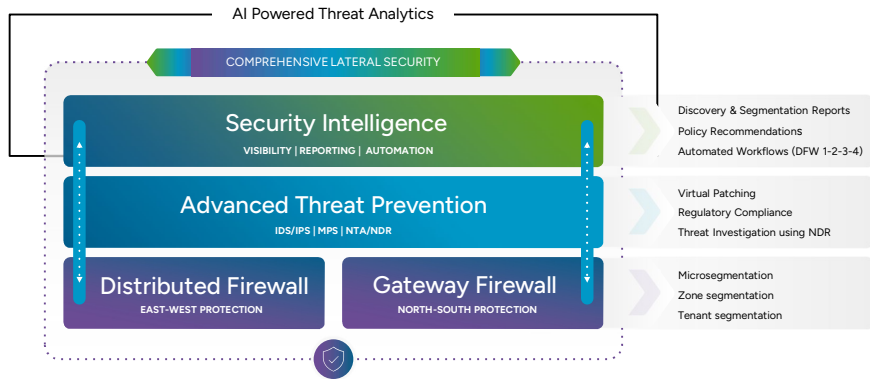
Limit Lateral Movement of Attackers

Network Access Control



Lateral Security, Deep Visibility, and Automated Workflows

VMware vDefend



Limit Lateral Movement of Attackers

Network Access Control with VMware Cloud Foundation 9

1. Add layers to public applications and websites with Avi and vDefend IDS/IPS/NTA.
2. **Egress rules, including DNS filtering and outbound inspection using vDefend Firewall.**
3. Identity-based microsegmentation between workloads with vDefend.
4. **Strict controls between infrastructure management interfaces and the rest of the organization. Consider privileged access workstations and out-of-band management.**
5. **Severely restrict access to ESX. Drive all access through vCenter & RBAC model.**
6. Canaries, honeypots, and tripwires on the network.
7. **Let vDefend automatically block problems when they are detected.**

See What Attackers Are Doing

Logging & Detection



Logs

ANALYZE LOGS
COMPARE LOGS

LOG SOURCES

Choose Saved Query

Search logs for text containing

PARTITION: All Partitions

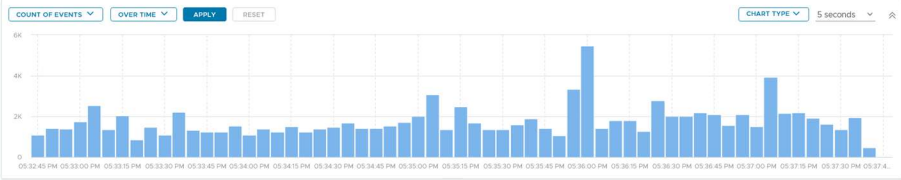
SM
1H
6H
24H
7D
CUSTOM

4/30/26 5:32 PM - 4/30/26 5:37 PM

+ ADD FILTER

COMPARE

SAVE QUERY



Stream
Event Types
Event Trends

Sorting By: Newest First

1 - 50 of 105.91K items

	Timestamp	Log
>	2026-04-30T22:37:38.312Z -05:00	2026-04-30T22:37:38.312Z INFO blueprint-service-app [host="tango-blueprint-service-app-5c4c75bc98-ks5zj" thread="generalScheduler-1" user="orgs=" trace=""] [bp=" proj=" dep=" req=" flow=" task=" tile=" res=" op="] c.v.t.b.telemetryLogMetricsConfig - threadPoolTaskExecutor executor stats: poolSize=0 active=0 corePoolSize=15 max=2147483647 prefix=tasks-
>	2026-04-30T22:37:38.312Z -05:00	2026-04-30T22:37:38.312Z INFO blueprint-service-app [host="tango-blueprint-service-app-5c4c75bc98-ks5zj" thread="generalScheduler-1" user="

Fields

FIELDS LIBRARY

Search Fields

app
cluster
component

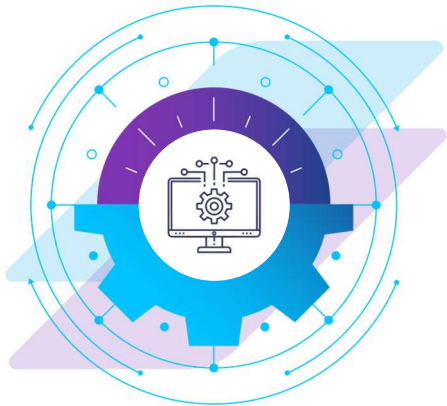
See What Attackers Are Doing

Logging & Detection with VMware Cloud Foundation 9

1. **Follow the VMware VCF Security Configuration Guide to enable audit logging everywhere.**
2. **Attackers are often using stolen credentials and have administrator access to the breached systems. Use VCF Log Management to aggregate, filter, and forward logs to your SIEM, out of reach from the attackers who might have access to the systems generating the data.**
3. Retain logs long enough to investigate a breach that started three years ago. This includes access logs for your IdP.
4. **Watch for legitimate operations performed by the wrong thing. Cloning by something that does not clone. Snapshots outside backup windows. Exports from accounts that do not export. See the VMware guidance on defending against BRICKSTORM for more info.**
5. Watch for changes to logging and audit configuration itself.

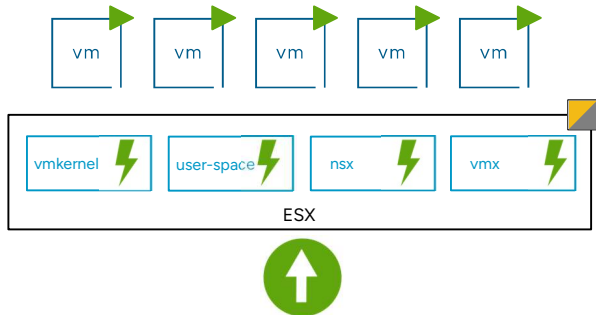
Patch Everything, Survive Anything

De-Risk Patching



Zero Downtime Updates At The Core of VCF

Live Patch of ESX vmkernel, VM runtime, NSX, System Daemons, and vCenter!



Most critical patching intended to incur minimal to no disruption

TPM-enabled hosts now fully supported for live patching

Some limitations apply: Confidential Computing, PodVM, skipping versions, big updates.

Purpose-Built Cyber Recovery With End-to-End Automation

DR and immutable backups are **not enough** for cyber recovery

AI-powered cyberattacks have **redefined cyber recovery requirements**

AI threats **exploit gaps** created by fragmented operations

1

ISOLATED CLEAN ROOM

Safe environment to power on workloads for validation before restore to production

2

VM NETWORK ISOLATION

Isolation of workloads during validation to prevent malware spread and reinfection

VMware Advanced Cyber Compliance

3

AI/ML-POWERED VALIDATION

Detection of malware-free, "living-off-the-land" attacks that remain undetected with traditional antivirus

4

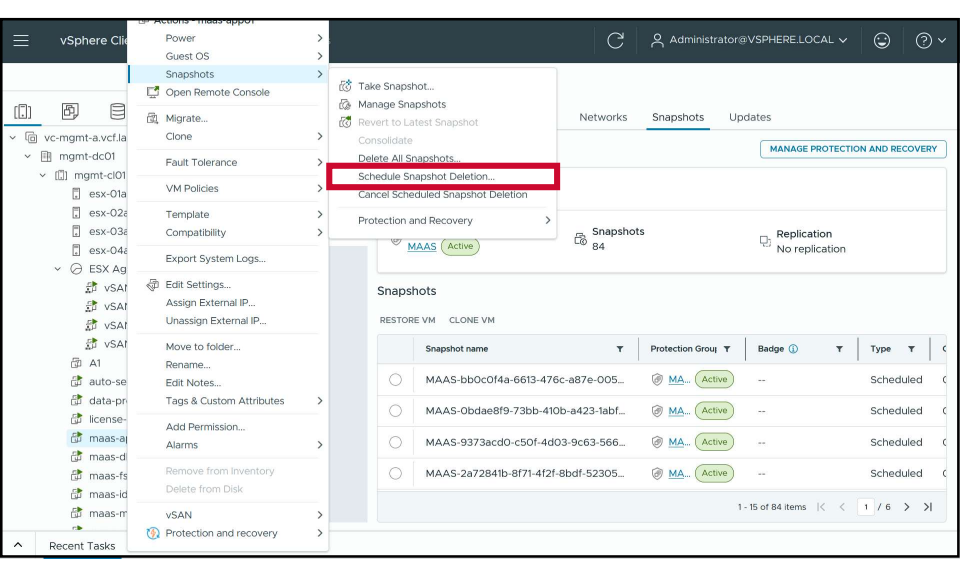
END-TO-END CYBER RECOVERY WORKFLOW

Fully-integrated cyber recovery to on-premises VCF sites with guided workflow automation

Make Rollback Easy Enough to Patch Anything

Workload Protection with VMware Cloud Foundation 9

1. **Software updates sometimes introduce unexpected change.**
The fix is a pipeline that catches them early, not a process that patches slowly.
2. **Snapshot before patching.** Automated through API, with automatic cleanup after a defined window (schedule snapshot cleanup).
3. **Deploy and use the vSAN Data Protection appliance for workload-level point-in-time recovery.**
4. **Use Avi or load balancer rolling updates to keep services up during patching.**
5. **Replicate critical workloads with VCF Cyber Recovery.** Use the testing tools to practice the recovery regularly so you know that it works, and how it will work.
6. **Make workloads start automatically when their VMs start.**



- Power
- Guest OS
- Snapshots
- Open Remote Console
- Migrate...
- Clone
- Fault Tolerance
- VM Policies
- Template
- Compatibility
- Export System Logs...
- Edit Settings...
- Assign External IP...
- Unassign External IP...
- Move to folder...
- Rename...
- Edit Notes...
- Tags & Custom Attributes
- Add Permission...
- Alarms
- Remove from Inventory
- Delete from Disk
- vSAN
- Protection and recovery

- Take Snapshot...
- Manage Snapshots
- Revert to Latest Snapshot
- Consolidate
- Delete All Snapshots...
- Schedule Snapshot Deletion...
- Cancel Scheduled Snapshot Deletion
- Protection and Recovery

Administrator@VSPHERE.LOCAL

Networks Snapshots Updates

MANAGE PROTECTION AND RECOVERY

Snapshots 84 Replication No replication

Snapshots

RESTORE VM CLONE VM

	Snapshot name	Protection Group	Badge	Type
	MAAS-bb0c0f4a-6613-476c-a87e-005...	MAAS Active	--	Scheduled
	MAAS-0bdae8f9-73bb-410b-a423-1abf...	MAAS Active	--	Scheduled
	MAAS-9373acd0-c50f-4d03-9c63-566...	MAAS Active	--	Scheduled
	MAAS-2a72841b-87f1-4f2f-8bdf-52305...	MAAS Active	--	Scheduled

1 - 15 of 84 items 1 / 6

Deletion Settings

Specify which snapshots you would like to delete. This will consolidate and remove these snapshots for this virtual machine. The snapshots will be consolidated to a single disk.

☐ Delete all snapshots

• Delete snapshots older than 3 days

CANCEL

[BACK](#)

SCHEDULE THE TASK

No items found

```
PowerShell
PS C:\Users\plankers> Get-VM | New-Snapshot -Name PrePatch

Name                Description                PowerState
----                -
PrePatch            PoweredOff
PrePatch            PoweredOff
PrePatch            PoweredOff
PrePatch            PoweredOff
PrePatch            PoweredOff
PrePatch            PoweredOff
PrePatch            PoweredOff
PrePatch            PoweredOff
PrePatch            PoweredOff
PrePatch            PoweredOff
PrePatch            PoweredOff
PrePatch            PoweredOff

New-Snapshot: 6/8/2026 4:51:31 PM      New-Snapshot      The operation for the entity "nsx-01a" f
ailed with the following message: "An error occurred while saving the snapshot: Exceeded the maximum num
ber of permitted snapshots.". An error occurred while saving the snapshot: Exceeded the maximum number o
f permitted snapshots. An error occurred while taking a snapshot: Exceeded the maximum number of permitt
ed snapshots.

New-Snapshot: 6/8/2026 4:51:31 PM      New-Snapshot      The operation for the entity "platform-s
ervices-a-hng7n" failed with the following message: "An error occurred while saving the snapshot: Exceed
ed the maximum number of permitted snapshots.". An error occurred while saving the snapshot: Exceeded th
e maximum number of permitted snapshots. An error occurred while taking a snapshot: Exceeded the maximum
number of permitted snapshots.

PrePatch            PoweredOff
PrePatch            PoweredOff
PrePatch            PoweredOff
PrePatch            PoweredOff
PrePatch            PoweredOff
```

```
PowerShell
PS C:\Users\plankers> Get-VM | Get-Snapshot | Remove-Snapshot

Confirm
Are you sure you want to perform this action?
Performing the operation "Removing snapshot." on target "VirtualMachineSnapshot-snapshot-2127".
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "Y"): A
PS C:\Users\plankers> |
```



maas-app01



ACTIONS

Summary

Monitor

Configure

Permissions

Datastores

Networks

Snapshots

Updates

Snapshot Management

Snapshot Deletion

Protection and Recovery

Snapshot Management

Protection and Recovery

MANAGE PROTECTION AND RECOVERY

Overview

Protection Groups



MAAS

Active



Snapshots

84



Replication

No replication

Snapshots

RESTORE VM

CLONE VM

	Snapshot name	Protection Group	Badge	Type	Created At	Expires At
	MAAS-bb0c0f4a-6613-476c-a87e-005...	MAAS Active	--	Scheduled	06/08/2026, 3:13:35 PM	06/15/2026, 3:13:35 PM
	MAAS-0bdæ8f9-73bb-410b-a423-1abf...	MAAS Active	--	Scheduled	06/08/2026, 1:13:35 PM	06/15/2026, 1:13:35 PM
	MAAS-9373acd0-c50f-4d03-9c63-566...	MAAS Active	--	Scheduled	06/08/2026, 11:13:35 AM	06/15/2026, 11:13:35 AM
	MAAS-2a72841b-8f71-4f2f-8bdf-52305...	MAAS Active	--	Scheduled	06/08/2026, 9:13:35 AM	06/15/2026, 9:13:35 AM
	MAAS-54a12676-3e7c-424c-8d82-2a8...	MAAS Active	--	Scheduled	06/08/2026, 7:13:35 AM	06/15/2026, 7:13:35 AM
	MAAS-9bf66cf3-a368-480c-979e-b441...	MAAS Active	--	Scheduled	06/08/2026, 5:13:35 AM	06/15/2026, 5:13:35 AM
	MAAS-c43771ce-7848-4fd5-8fc3-bb78...	MAAS Active	--	Scheduled	06/08/2026, 3:13:35 AM	06/15/2026, 3:13:35 AM
	MAAS-cd7be1e6-3f8d-49df-837f-0d56...	MAAS Active	--	Scheduled	06/08/2026, 1:13:35 AM	06/15/2026, 1:13:35 AM

1 - 15 of 84 items

1

/ 6

>

>

Recent Tasks

Alarms

📁 mgmt-cl01 ⋮ ACTIONS

- Summary
- Monitor
- Configure**
- Permissions
- Hosts
- VMs
- Datastores
- Networks
- Updates

- Services** ▾
 - vSphere DRS
 - vSphere Availability
- Configuration** ▾
 - Quickstart
 - General
 - Key Provider
 - VMware EVC
 - VM/Host Groups
 - VM/Host Rules
 - VM Overrides
 - I/O Filters
 - Host Options
 - Host Profile
- Licensing
- Alarm Definitions
- Scheduled Tasks
- Desired State** ▾
 - Configuration
- vSAN ▾

Protection and Recovery

Use Protection and Recovery to protect your VMs with local and remote vSAN snapshots against disasters, outages, and cyber threats.

- SUMMARY**
- PROTECTION GROUPS
- VMS
- REPLICATION

Overview

🛡️ Protection groups
1

Active 1

30% Protected VMs

📁 Protected VMs
12 (out of 40) ⓘ

- VMs with local protection 12 (30%)
- Not protected VMs 28 (70%)

vSAN Snapshot Space Usage

939.99 GB (18.56%) Free usable capacity 4.03 TB

Used capacity 939.99 GB (18.56%) ⚙️ Used by vSAN snapshots 25.22 GB (0.49%)

ⓘ Scheduled snapshots will not be taken if the datastore's capacity exceeds 70% threshold.

[VIEW vSAN CAPACITY](#)

The investments in process
improvement that make patching safe
also make **ransomware recovery,**
hardware failure, application upgrades,
and human error survivable.

Security Hardening & Compliance Resources

<https://brcm.tech/vcf-security>

<https://github.com/vmware/vcf-security-and-compliance-guidelines/>



Baseline Hardening Made Easy Across All VCF Components

VCF 9.1 Security Configuration Guide at <https://bcm.tech/vcf-security>

VMware Cloud Foundation Security Configuration Guide									
Version 910-20200612-01									
SCG ID	Secure Controls Framework ID	DISA STIG ID	PCIDSS 4.0.1 ID	NIST 800-53RS ID	Component Name	Component Version	Feature/Function	Description/Title	Discussion
automation-9.activity-log	MON-10, CFG-03	N/A	10.5.1	AU-11, CM-07	Automation	9.1.0	Base	Retain the VCF Automation activity log for the audit window and keep debug output disabled.	VCF Automation's activity through the platform's changes made through days entries are kept by investigations and audit before anyone looks to only how far back the log of debug information is left disabled, as it ships read it.
automation-9.audit-settings	IAC-10, CRY-03	N/A	4.2.1	IA-05, SC-08	Automation	9.1.0	Base	Bound VCF Automation OIDC token lifetimes and keep redirect targets HTTPS-only.	Not applicable if VCF A When VCF Automation identity provider, these callback URLs it will re issued access token is: lifetimes used elsewhere outside its peers. The re to obtain new access to means a stolen refresh redirect URL, policy co restricting it to HTTPS-c being sent to an unenc
automation-9.operation-limits	CAP-02	VCFA-91-000104	N/A	SC-05, SC-06	Automation	9.1.0	Base	Configure operation limits in VCF Automation to bound resource-intensive operations.	VCF Automation lets be deletions, large catalog operation limits, a risk

Filterable based on component, feature, version, and risk

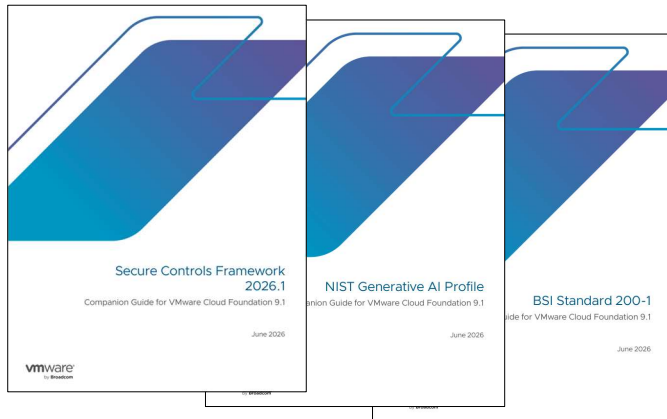
Standardizes checks across components

Maps additional regulations and differences

Guidance that is easy to use and understand for VCF components

Understand How VCF Features Meet Regulatory Requirements

VCF Compliance Companion Library at <https://brcm.tech/vcf-security>



27 standards + the Secure Controls Framework itself

XLSX, CSV, Markdown, PDF

Know which VCF features apply to a specific regulatory requirement.

Aids auditor understanding of VCF features.

Security & Compliance Repository: brcm.tech/vcf-security

<https://github.com/vmware/vcf-security-and-compliance-guidelines>

vmware / vcf-security-and-compliance-guidelines

Code Issues Pull requests Actions Security Insights

vcf-security-and-compliance-guidelines Public

Edit file Unwatch 27 Fork Starred 110

main 1 branch 0 tags

Go to file Add file Code

planckers Updated ransomware defense PDFs to resolve MacOS compatibility 11ea57c · last week 216 Commits

features-capabilities	Audit Event References for vSphere & Cloud Foundation	2 weeks ago
ransomware-resources	Updated ransomware defense PDFs to resolve MacOS comp...	last week
regulatory-compliance	Minor updates to VCF/TCP UK TSA CoP guidance.	last month
security-advisories	Back-in-time issues resolved with the release of vSphere BU3...	6 months ago
security-configuration-hardening-guide	VCF SCG 901-20260107-01 & 803-20260107-01	2 weeks ago
security-design	Tips & Tricks for Upgrading vSphere	8 months ago
.gitignore	Initial content migration.	2 years ago
CODE_OF_CONDUCT.md	Initial content migration.	2 years ago
CONTRIBUTING_CLA.md	Initial content migration.	2 years ago
LICENSE.md	Update and rename LICENSE to LICENSE.md	2 years ago
LINKS.md	KB link for diagnostic data and support dumps.	9 months ago
NOTICE.md	Rename NOTICE to NOTICE.md	2 years ago

About

Security, compliance, and operational resilience resources applicable to VMware Cloud Foundation and VMware vSphere. This repository is an official VMware repository managed by Broadcom staff.

[brcm.tech/vcf-security](#)

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Contributors

SCG & sample code
for automation

Regulatory
compliance library

Whitepapers on
standards and
design

Q&A for product
capabilities

"One-stop shop" for
platform security



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Thank You

